

Integers

Integers

“God made the integers; all else is the work of man”

Leopold Kroenecker

Division

- *Most of useful properties of integers are related to division*
- *If a and b are integers with $a \neq 0$, we say that a **divides** b if there is an integer c such that $b = ac$.*
- *When a divides b we say that a is a **divisor** (**factor**) of b , and that b is a **multiple** of a .*
- *The notation $a \mid b$ denotes that a divides b . We write $a \nmid b$ when a does not divide b*
- *Example. Let n and d be positive integers. How many positive integers not exceeding n are divisible by d ?*

The numbers in question have the form dk , where k is a positive integer and $0 < dk \leq n$. Therefore, $0 < k \leq n/d$. Thus the answer is $\lfloor n/d \rfloor$

Properties of Divisibility

● Let a , b , and c be integers. Then

(i) if $a \mid b$ and $a \mid c$, then $a \mid (b + c)$;

(ii) if $a \mid b$, then $a \mid bc$ for all integers c ;

(iii) if $a \mid b$ and $b \mid c$, then $a \mid c$.

● *Proof.*

(i) Suppose $a \mid b$ and $a \mid c$. This means that there are k and m such that $b = ak$ and $c = am$.

Then $b + c = ak + am = a(k + m)$, and a divides $b + c$.

Properties of Divisibility (cntd)

● If a , b , and c are integers such that $a \mid b$ and $a \mid c$, then $a \mid mb + nc$ whenever m and n are integers.

● *Proof.*

By part (ii) it follows that $a \mid mb$ and $a \mid nc$.

By part (i) it follows that $a \mid mb + nc$.

● If $a \mid b$ and $b \mid a$, then $a = \pm b$.

● *Proof.*

Suppose that $a \mid b$ and $b \mid a$. Then $b = ak$ and $a = bm$ for some integers k and m .

Therefore $a = bm = akm$, which is possible only if $k, m = \pm 1$.

Euclidean division algorithm

● Theorem (*Euclidean division*)

Let a be an integer and d a positive integer. Then there are unique integers q and r , with $0 \leq r < d$, such that $a = dq + r$

● d is called the **divisor**, a is called the **dividend**, q is called the **quotient**, and r is called the **remainder**

● Examples:

- $a = 101$ and $d = 11$

- $a = -11$ and $d = 3$

- $a = 3$ and $d = 11$

Representation of Integers

- *In most cases we use decimal representation of integers. For example, 657 means*

$$6 \cdot 100 + 5 \cdot 10 + 7 = 6 \cdot 10^2 + 5 \cdot 10^1 + 7 \cdot 10^0$$

- *Let b be a positive integer greater than 1. Then if n is a positive integer, it can be expressed uniquely in the form*

$$n = a_k b^k + a_{k-1} b^{k-1} + \cdots + a_1 b + a_0$$

where k is a nonnegative number, $a_k, a_{k-1}, \dots, a_1, a_0$ are nonnegative integers less than b , and $a_k \neq 0$

- *Such a representation of n is called the **base b expansion of n** , denoted by $(a_k a_{k-1} \dots a_1 a_0)_b$*

Binary Expansion

- Important case of a base is 2. The base 2 expansion is called the binary expansion of a number

$$n = a_k \cdot 2^k + a_{k-1} \cdot 2^{k-1} + \cdots + a_1 \cdot 2 + a_0$$

- Find the binary expansion of 165

							a_7
							①
$2 \overline{) 165}$	$2 \overline{) 82}$	$2 \overline{) 41}$	$2 \overline{) 20}$	$2 \overline{) 10}$	$2 \overline{) 5}$	$2 \overline{) 2}$	$2 \overline{) 1}$
①	①	①	①	①	①	①	①
a_0	a_1	a_2	a_3	a_4	a_5	a_6	

$$165 = 2 \cdot 82 + 1$$

$$20 = 2 \cdot 10 + 0$$

$$2 = 2 \cdot 1 + 0$$

$$82 = 2 \cdot 41 + 0$$

$$10 = 2 \cdot 5 + 0$$

$$1 = 2 \cdot 0 + 1$$

$$41 = 2 \cdot 20 + 1$$

$$5 = 2 \cdot 2 + 1$$

$$165 = (10100101)_2$$

Hexadecimal Expansion

● *Using A, B, C, D, E, F for 10, 11, 12, 13, 14, 15, respectively, find the base 16 expansion of :*

- 174

- 218

Primes

- *Every integer n (except for 1 and -1) has at least 2 positive divisors, 1 and n itself (or $-n$).*
- *A positive number that does not have any other positive divisor is called **prime**.*
- *Prime numbers: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, ...*
- *Mersenne numbers are numbers of the form $M_n = 2^n - 1$*
- *There are many prime numbers among Mersenne numbers. The greatest known prime number is $M_{77232917} = 2^{77232917} - 1$*
- *The next candidate is $M_{M_{61}} = 2^{2305843009213693951} - 1$*
- *A positive number that is not prime is called **composite***

Composite Numbers

● *Every composite number has a prime divisor.*

● *Proof.*

Let S be the set of all composite numbers that do not have a prime divisor

Since $S \subseteq \mathbb{N}$, by the Well-Ordering Principle, it has a least element r .

As r is not prime, it has a divisor, therefore, $r = uv$ for some positive integers u and v .

$u < r$ and $v < r$. Therefore $u \notin S$, and u has a prime divisor p .

Since $p \mid u$ and $u \mid r$, we conclude that $p \mid r$, a contradiction.

How many prime numbers are there?

● Theorem (Euclid)

There are infinitely many prime numbers.

● Proof.

By contradiction. Suppose that $\{p_1, p_2, \dots, p_k\}$ is the set of all prime numbers, and let $a = p_1 p_2 \dots p_k + 1$

Since a is greater than any member of the list, a is composite.

By the previous statement, a has a prime divisor, that is for some p_j we have $p_j \mid a$

Since $p_j \mid a$ and $p_j \mid p_1 p_2 \dots p_k$ we have $p_j \mid a - p_1 p_2 \dots p_k = 1$

A contradiction.

● If n is a positive integer, then there are approximately $\frac{n}{\ln n}$ prime numbers not exceeding n

Open Problems about Primes

● Goldbach's Conjecture

*Every positive **even** number can be represented as the sum of two prime numbers.*

For example: $4 = 2 + 2$, $8 = 5 + 3$, $42 = 37 + 5$

Goldbach's conjecture is known to be true for even numbers up to $4 \cdot 10^{18}$

● The Twin Prime Conjecture

Twin primes are primes that differ by 2, such as 3 and 5, 5 and 7, 11 and 13, etc.

The Twin Prime Conjecture asserts that there are infinitely many twin primes.

The record twin primes: $2,996,863,034,895 \cdot 2^{1,290,000} \pm 1$

Practice

Exercises from the Book

7th edition: 6, 8, 23 (page 244 – 245)

1ab, 3ac, 7ac (also find the decimal representation) (page 255)

8th edition: 6, 8, 29 (page 258 – 259)

1ab, 3ac, 7ac (also find the decimal representation) (page 269)